

Skyrmion Cyclotron Resonance in Ferrimagnets

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We show that a resonance due to gyroscopic motion of skyrmions, conceptually similar to the electron cyclotron resonance in metals, can be excited in a ferrimagnetic film by a spin current or microwaves. It must permit unambiguous measurement of the skyrmion mass for which a universal expression depending solely on the exchange interaction between spins belonging to two different ferrimagnetic sublattices is derived. The dependence of the skyrmion cyclotron frequency on parameters is computed for a TM/RE ferrimagnet, using CoGd as an example. The cyclotron frequency exhibits a dip near the angular momentum compensation point, where it hybridizes with the ferromagnetic resonance. The skyrmion cyclotron mode is studied for individual skyrmions and for skyrmion lattices, where the effect must be strong enough to be observed in microwave and spin-current experiments.

In this Letter, we address two interconnected problems that are at the forefront of theoretical and experimental research on magnetism of solids: spin excitations in ferrimagnets near the angular momentum compensation point, and the mass of a ferrimagnetic skyrmion. Recent interest to ferrimagnets has been generated by observation that while possessing a nonzero magnetization they can exhibit fast dynamics similar to that of compensated antiferromagnets [1–16]. It stems from the fact that non-equal gyromagnetic factors g_i of antiferromagnetically coupled, ferromagnetically ordered sublattices permit a nonzero average magnetic moment, $\sum_i g_i \mu_B \langle \mathbf{S}_i \rangle$ when the average angular momentum $\sum_i \hbar \langle \mathbf{S}_i \rangle$ associated with the atomic spins of the sublattices $\mathbf{S}_i$ turns zero at a certain temperature or composition of the ferrimagnet. Hardening of the magnetic resonance, and spin waves in general, on approaching the angular momentum compensation point in ferrimagnets has been demonstrated by several theoretical studies, see, e.g., 17–23, and confirmed by numerous experiments, see, e.g., 1, 6, 20, 24, and 25.

Likewise, the interest in skyrmions, besides their mathematical beauty, has been inspired by their potential for topologically protected information technology [26]. Various inhomogeneous magnetic phases, including magnetic domains and skyrmion lattices (SkL), have been observed in ferrimagnetic films and racetracks [27–30]. Excitation modes of individual skyrmions and SkL have been studied in ferromagnets (see, e.g., Refs. 31 and 32, and references therein), but not so much in ferrimagnets. The dynamics of skyrmions largely depends on whether it is massive or massless. The question of the skyrmion mass has been one of the most controversial in skyrmion physics [33–39]. It has been rigorously demonstrated that in a 2D ferromagnetic film the inertial mass associated with the motion of the center of the topological charge of the skyrmion is zero [40, 41]. It was suggested that a nonzero skyrmion mass can arise from its confinement in a nanoring [42] or nanotrack [41], or it can be generated by the interaction with other degrees of freedom, such as

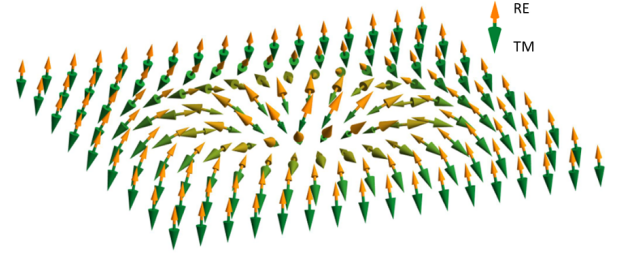


Figure 1. Two-sublattice skyrmion in a TM/RE ferrimagnet. Far from the skyrmion’s core, TM spins are directed down (green) and RE spins are directed up (orange).

phonons [43].

In this Letter we show that a finite skyrmion mass is generated in a ferrimagnet by the inter-sublattice exchange interaction alone. The mass is independent of the atomic spins and spin densities of the sublattices. The finite mass of the skyrmion generates its gyroscopic motion at a frequency $\Omega = (J'/\hbar)(S - c\Sigma)$, where $J' > 0$ is the strength of the antiferromagnetic exchange interaction between spins S and Σ of the sublattices in $\mathcal{H}_{\text{int}} = J' \sum_i \mathbf{S}_i \cdot \mathbf{\Sigma}_i$, and c is the concentration of spins Σ . We call Ω the frequency of the skyrmion cyclotron resonance (SCR), and confirm its analytical value by numerical computation of spin-current and microwave power absorption that uses a microscopic lattice-spin model. The SCR frequency decreases on approaching the angular momentum compensation point $c \rightarrow c^* = S/\Sigma$, where it hybridizes with the FMR frequency, pushing it down. (Having a TM/RE ferrimagnet in mind we assume $S < \Sigma$). Our calculations have been performed for individual skyrmions and SkL using parameters of a CoGd ferrimagnet, for which our predictions can be tested in experiments. Measurements of the SCR must allow unambiguous determination of the evasive skyrmion mass, similar to how the effective electron mass in metals was obtained from measurements of the Azbel-Kaner electron

cyclotron resonance [44].

A ferrimagnetic skyrmion is shown in Fig. 1. A classical 2D field-model counterpart of $\mathcal{H}_{\text{int}} = J' \sum_i \mathbf{S}_i \cdot \mathbf{\Sigma}_i$ on a square lattice of spacing a , suitable for its study, is $U_{\text{int}} = J'a^2 \int d^2r \rho_S(\mathbf{r}) \cdot \rho_\Sigma(\mathbf{r})$, where ρ_S and ρ_Σ are spin densities of the sublattices $\rho_S = S/a^2$ and $\rho_\Sigma = c\Sigma/a^2$ respectively. We assume [45–47] that they are given by rigid skyrmion solutions $\rho_S = \rho_S \mathbf{f}(\mathbf{r} - \mathbf{R}_S)$ and $\rho_\Sigma = \rho_\Sigma \mathbf{f}(\mathbf{r} - \mathbf{R}_\Sigma)$ centered at $\mathbf{R}_S$ and $\mathbf{R}_\Sigma$ respectively, separated by the distance $\mathbf{d} = \mathbf{R}_S - \mathbf{R}_\Sigma$, which is very small compared to the size of the skyrmion λ . This yields

$$U_{\text{int}} = J'a^2 \rho_S \rho_\Sigma \int d^2r \mathbf{f}(\mathbf{r}) \cdot \mathbf{f}(\mathbf{r} + \mathbf{d}) \cong \frac{1}{2} J'a^2 \mathcal{D} \rho_S \rho_\Sigma \mathbf{d}^2, \quad (1)$$

where $\mathcal{D} = \frac{1}{2} \int dxdy [\partial \mathbf{f}(\mathbf{r}) / \partial \mathbf{r}]^2$ equals $4\pi|Q|$ for a Belavin-Polyakov pure-exchange skyrmion [48] with a topological charge Q . Notice that the exchange energy associated with the separation of the centers of sublattice skyrmions scales as $J'(d/a)^2$, so that in practice one should always expect $d \ll a$.

In the absence of external forces, the dynamics of the two skyrmions moving at velocities $\mathbf{V}_S$ and $\mathbf{V}_\Sigma$ satisfies Thiele equations [49] augmented by the spin-current terms

$$\begin{aligned} -\mathbf{G}_S \times (\mathbf{V}_S - \mathbf{v}_{s,S}) &= \mathbf{F}_S + \mathbf{F}_{S\Sigma}, \\ \mathbf{G}_\Sigma \times (\mathbf{V}_\Sigma - \mathbf{v}_{s,\Sigma}) &= \mathbf{F}_\Sigma + \mathbf{F}_{\Sigma S}, \end{aligned} \quad (2)$$

where $\mathbf{G}_{S,\Sigma} = 4\pi\hbar|Q|\rho_{S,\Sigma}\mathbf{e}_z$ are the so-called gyrovec-tors, $\mathbf{v}_{s,S}$ and $\mathbf{v}_{s,\Sigma}$ are spin currents in the TM and RE sublattices, $\mathbf{F}_S$ and $\mathbf{F}_\Sigma$ are external forces acting on the S - and Σ -skyrmions, while the interaction force is given by $\mathbf{F}_{S\Sigma} = -\mathbf{F}_{\Sigma S} = -\partial U_{\text{int}} / \partial \mathbf{R}_S = \partial U_{\text{int}} / \partial \mathbf{R}_\Sigma = -J'a^2 \mathcal{D} \rho_S \rho_\Sigma \mathbf{d}$. In the simplest case without external forces and spin currents, for the velocity $\mathbf{V} = \dot{\mathbf{R}}$ of the coordinate of the center of mass of the two skyrmions, $\mathbf{R} = (\mathbf{R}_S + \mathbf{R}_\Sigma)/2$, and the acceleration $\dot{\mathbf{V}}$ from Eq. (2) resolved for the velocities one obtains

$$\mathbf{V} = -\frac{1}{2} J'a^2 \mathcal{D} \rho_S \rho_\Sigma \left(\frac{1}{G_S} + \frac{1}{G_\Sigma} \right) \mathbf{e}_z \times \mathbf{d}, \quad (3)$$

$$\dot{\mathbf{V}} = -J'a^2 \mathcal{D} \rho_S \rho_\Sigma \left(\frac{1}{G_S} - \frac{1}{G_\Sigma} \right) \mathbf{e}_z \times \mathbf{V}. \quad (4)$$

The second equation describing the motion of the free skyrmion is obtained by the differentiation of the first one, expressing $\dot{\mathbf{d}}$ through the velocities, expressing the latter through $\mathbf{d}$ again using Eq. (2), and using the first equation to eliminate $\mathbf{d}$. This equation can be rewritten as $\dot{\mathbf{V}} = \Omega \times \mathbf{V}$, where the cyclotron frequency of the skyrmion is given by a very simple expression

$$\Omega = \frac{J'}{\hbar} \frac{\mathcal{D}}{4\pi|Q|} (S - c\Sigma) \mathbf{e}_z \Rightarrow \frac{J'}{\hbar} (S - c\Sigma) \mathbf{e}_z, \quad (5)$$

if $\mathcal{D} = 4\pi|Q|$. The corresponding cyclotron radius is $R_c = V/|\Omega|$ and diverges at the angular-momentum

compensation point. From Eq. (3) one obtains $V = (1/2)(J'/\hbar)\mathcal{D}/(4\pi|Q|)(S + c\Sigma)d$, thus $R_c = (d/2)(S + c\Sigma)/|S - c\Sigma|$. As d is very small, one needs to come close to the angular-momentum compensation point to have a significant R_c .

To identify the skyrmion mass, one should include the external force $\mathbf{F} = \mathbf{F}_S + \mathbf{F}_\Sigma$. With a slightly more effort, for $\mathbf{F} = \text{const}$ one obtains

$$M\dot{\mathbf{V}} = (\mathbf{G}_S - \mathbf{G}_\Sigma) \times \mathbf{V} + \mathbf{F} = M\Omega \times \mathbf{V} + \mathbf{F}, \quad (6)$$

where $M = (4\pi\hbar Q)^2 / (J'a^2 \mathcal{D})$ is the skyrmion mass. For $D = 4\pi|Q|$ it equals

$$M = \frac{4\pi\hbar^2|Q|}{J'a^2}. \quad (7)$$

It must be of the order of the proton mass for a single 2D atomic layer of CoGd ferrimagnet. Experimentally, the skyrmion mass can be found as

$$M = \frac{G_S - G_\Sigma}{\Omega} \quad (8)$$

from the cyclotron-resonance experiment driven by the spin current or microwave radiation, where the difference of gyrovec-tors $G_S - G_\Sigma$ plays the role of the magnetic field acting on an electric charge. Remarkably, M doesn't depend on the spins (cf. Ref. [46]). The proportionality of the skyrmion mass to $|Q|$ points to its topological origin. One should keep in mind, however, that the applicability of Eq. (7) depends on two assumptions. The first is the condition of the rigidity of the sublattice skyrmions, which requires the exchange interaction J responsible for the skyrmion texture within each ferromagnetically ordered sublattice to be large compared to the intersub-lattice exchange interaction J' . The second is that the skyrmion is formed by many spins belonging to each sublattice, which requires $c \gg (a/\lambda)^2$, where λ is the size of the skyrmion.

Equation (6) should be augmented by the inhomogeneous terms due to spin current and time derivatives of the external force and the spin current. The addition of weak dissipation to the Thiele equations (2) adds a small imaginary part, $i\Lambda(J'/\hbar)(S + c\Sigma)$, to Eq. (5), where $\Lambda \equiv \alpha\mathcal{D}/(4\pi|Q|)$ and $\alpha \ll 1$ is the phenomenological damping constant in the Landau-Lifshitz equation for the magnetization. The mass gets modified by a factor $1 + \Lambda^2$.

Notice that the lattice spacing a that we used in the definitions of spin densities and gyrovec-tors has canceled out from the formula for Ω . For a film consisting of N atomic layers, both M and the gyrovec-tors become proportional to N . Consequently, the expression (5) for the skyrmion cyclotron frequency, derived for a single 2D layer, remains unchanged in a film of finite thickness.

To test our analytical theory, we studied numerically excitations in a 2D model of a TM/RE ferrimagnet with

skyrmions, described by the Hamiltonian

$$\begin{aligned} \mathcal{H} = & -\frac{1}{2} \sum_{ij} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + J' \sum_i \mathbf{S}_i \cdot p_i \boldsymbol{\Sigma}_i - \frac{D}{2} \sum_i S_{i,z}^2 \\ & + A \sum_i [(\mathbf{S}_i \times \mathbf{S}_{i+\delta_x})_x + \mathbf{S}_i \times \mathbf{S}_{i+\delta_y})_y]. \end{aligned} \quad (9)$$

Here J_{ij} is the ferromagnetic nearest-neighbor exchange within the square TM-sublattice of spins $\mathbf{S}_i$ with the coupling constant $J > 0$. The RE spins $\boldsymbol{\Sigma}_i$ occupy a similar square lattice and are coupled to the TM spins with the coupling constant $J' > 0$. These spins are diluted, which is taken into account by the occupation factors $p_i = 0, 1$, so that $\langle p_i \rangle = c$. Other terms are the easy-axis anisotropy and the Dzialoshinskii-Moriya interaction (DMI) within the TM sublattice. Having CoGd in mind, we chose $S = 3/2$, $\Sigma = 7/2$, $J'/J = 0.2$, $D/J = 0.03$, and $A/J = 0.1$. To the Hamiltonian above, one can add the Zeeman term due to the external magnetic field, as well as the time-dependent microwave (MW) field. In this model, Gd spins are slaves of the Co spins. Consequently, when the separation of the centers of Co and Gd skyrmions is small compared to the lattice spacing, one can still expect the condition of the rigidity of the sublattice skyrmions to be roughly satisfied. The dynamics of the lattice spins is described by the Landau-Lifshitz (LL) equation, augmented by the spin-current term:

$$\hbar \frac{\partial \mathbf{S}_i}{\partial t} + \hbar (\mathbf{v}_{s,S} \cdot \nabla) \mathbf{S}_i = \mathbf{S}_i \times \mathbf{H}_{\text{eff},S,i}, \quad (10)$$

where $\mathbf{H}_{\text{eff},S,i} = -\partial \mathcal{H} / \partial \mathbf{S}_i$ is the effective field (in the energy units) acting on the TM sublattice and $\mathbf{v}_{s,S}$ is the spin current acting on the TM spins $\mathbf{S}_i$. The action of this streaming term is translating the whole set of $\mathbf{S}_i$ with the velocity $\mathbf{v}_{s,S}$. The equation of motion for the RE spins is similar, only the spin-current term is negligible because RE spins are due to the f -electrons which are close to the atomic cores and effectively decoupled from the charge carriers. We do not include the phenomenological damping in the LL equation since the model shows a significant intrinsic damping due to the RE disorder and/or temperature.

In the absence of skyrmions, the ferrimagnet described by the above Hamiltonian possesses two excitation modes: the low-frequency acoustic mode and the high-frequency optical mode. Various limits for these modes have been worked out in Ref. 23. At $c \rightarrow 0$, the acoustic mode tends to the FMR frequency of the Co sublattice alone, $\hbar \omega_{\text{LF}} / (SJ) = D/J = 0.03$. As c increases, the frequency of this mode goes up. For $D \ll J'$, at the angular-momentum compensation, $c = S/\Sigma$, one has $\hbar \omega_{\text{HF,LF}} = S (\sqrt{DJ'} \pm D/2)$, making $\hbar \omega_{\text{LF}} / (SJ) \approx 0.0625$ and $\hbar \omega_{\text{HF}} / (SJ) \approx 0.0925$. At small c , the SCR frequency Ω of Eq. (5), divided by $SJ/\hbar$, is approximately $J'/J = 0.2$. In real units for CoGd ferrimagnet,

$\Omega / (2\pi) \Rightarrow SJ' / (2\pi\hbar)$ must be in the ballpark of a few hundred GHz at small c , going down on approach to compensation, $c \rightarrow S/\Sigma$, where it may be accessible by experiments. Since the frequency of the acoustic mode goes up on increasing c from zero, while the frequency of the cyclotron mode goes down, one should expect their hybridization on approach to the angular momentum compensation. This has been observed in the numerical experiment described below. Physically, the mode coupling occurs because precession of spins in the acoustic mode results in the circular motion of the skyrmion's top that is the cyclotron motion. For this reason, the latter can be excited by the MW radiation.

To find the equilibrium states of the system with a single skyrmion or a skyrmion lattice (SkL) at $T = 0$, we performed the energy minimization starting from the appropriate initial conditions (see, e.g., Ref. [50]). To find the modes' frequencies, we excited the system with the sinc spin current or a sinc MW field directed along the x axis (perpendicular to the spins in the bulk and in the skyrmions' cores), e.g., $\mathbf{v}_{s,S}(t) = \mathbf{e}_x v_s \sin(\omega_{\text{max}} t) / (\omega_{\text{max}} t)$. The Fourier spectrum of this function is a constant up to the cutoff frequency ω_{max} and zero above it, so that it excites all modes in the interval $0 < \omega < \omega_{\text{max}}$. The LL equation was solved with the 5th-order Butcher's Runge-Kutta method (RK5) that makes 6 function evaluations over one integration step. We used moderate integration times such as $t_{\text{max}} \approx 6000$ with $SJ \Rightarrow 1$ and $\hbar \Rightarrow 1$, as usual, and at the end computed the fluctuation spectrum (FS) of the quantity F as follows:

$$\text{FS}(\omega) = \frac{1}{t_{\text{max}}} \left| \int_0^{t_{\text{max}}} F(t) e^{i\omega t} dt \right|^2. \quad (11)$$

As the quantity F we used either the X displacement of the skyrmion (or the average displacement of skyrmions in the SkL) in the TM subsystem, X_S , from its equilibrium position or the x component of the magnetic moment of the system. The modes' frequencies were searched for as the resonances in $\text{FS}(\omega)$. We use the system size 116×132 in lattice units (with periodic boundary conditions) that neatly comprises an SkL of 12 skyrmions, as shown in Fig. 2.

The dependence of the hybridized SCR and LF ferrimagnetic modes on the concentration of Gd c for a single skyrmion (with the size $\lambda_{\text{eff}}/a \approx 18$) in the system in the case of spin-current excitation is shown in Fig. 3. As the result of hybridization, there are upper and lower modes with a gap between them. Far from the angular-momentum compensation (AMC) point, $c^* = S/\Sigma = 0.429$, the upper mode is the SCR mode, and the lower mode is the LF ferrimagnetic mode. In the region around the AMC point, the lower mode becomes mainly the SCR mode with the frequency going to zero

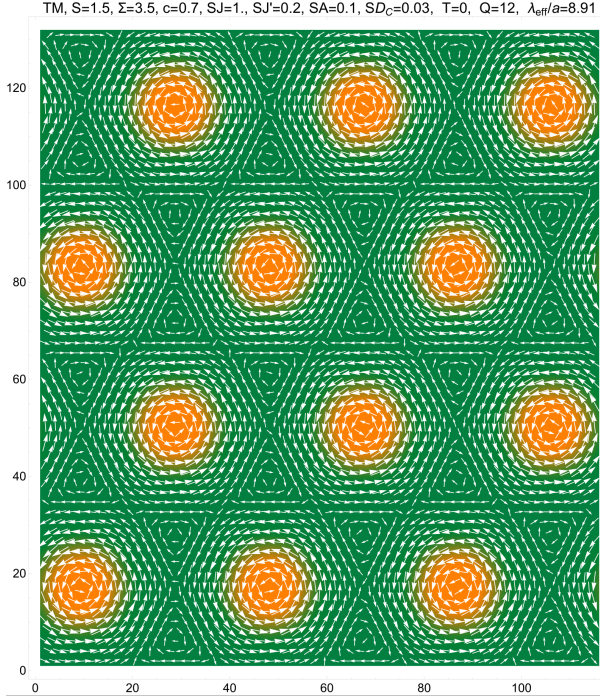


Figure 2. The system with a skyrmion lattice (TM spins are shown) for the chosen model parameters and $c = 0.7$. The skyrmion size $\lambda_{\text{eff}}/a \approx 8.9$ is defined according to Ref. [51].

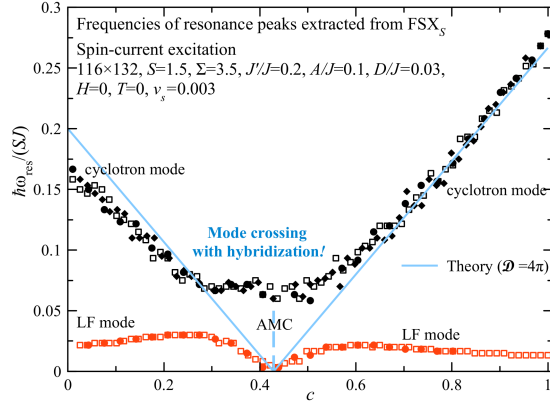


Figure 3. The dependence of the hybridized SCR and LF ferrimagnetic modes on the concentration of Gd in a model of CoGd ferrimagnet. (Single skyrmion in a 116×132 system, spin-current excitation, and the TM skyrmion position, X_S , as the observed quantity. The results for the SkL are similar).

at $c = c^*$. For small c , the peaks in $\text{FS}(\omega)$ corresponding to both upper and lower modes become very small and finally disappear in the limit $c \rightarrow 0$. We have separately checked the proportionality of Ω to J' at a constant c far from compensation. Excellent quantitative agreement with the analytical formula (5) is apparent, as is the expected qualitative dependence of both modes on c due to their hybridization.

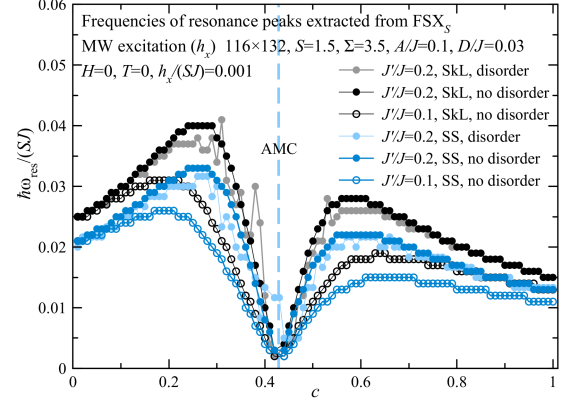


Figure 4. The dependence of the lower mode on the concentration of Gd for a single skyrmion (SS) and the skyrmion lattice in a model of CoGd ferrimagnet.

The frequencies of the modes can also be obtained by computing the energy absorbed at different frequencies for a given c . There are two absorption peaks corresponding to the upper and lower modes. With an eye on experiments, one should worry about the power absorbed by the cyclotron mode. A single skyrmion can hardly provide a detectable signal, so one should consider the power absorption by a dense skyrmion lattice created in a ferrimagnet. The modes' frequencies for the SkL obtained via $\text{FS}(\omega)$, are rather close to the results of Fig. 3.

A similar computation with the microwave excitation reproduces the lower mode, while the upper mode cannot be reliably resolved. In these experiments, the observed quantity F in Eq. (11) can be both the skyrmion displacement X_S and the transverse magnetization component. The latter is easier to detect experimentally. These observations point toward the possibility to obtain the evidence of the skyrmion cyclotron motion in the microwave experiment as well.

Figure 4 shows the RE concentration dependence of lower mode's frequency in the case of MW excitation for a single skyrmion (SS) and the SkL. In both cases, there is a spectacular depression near the AMC point. The curves labeled "no disorder" were obtained for the idealized model in which the length of each RE spin is $c\Sigma$. These curves do not have the scatter caused by the disorder. For smaller intersublattice coupling, $J'/J = 0.1$, the depression is expectedly broader.

In conclusion, we have demonstrated that a universal formula for the skyrmion mass in a ferrimagnet corresponds to a universal formula for the frequency of the gyroscopic motion of the skyrmion. Such a motion of skyrmions in ferrimagnets must show as a skyrmion cyclotron resonance, SCR, in the absorption of the power of the ac spin current, and in the hybridization with ferri-

magnetic modes when skyrmions are present, which must be possible to detect in microwave experiments as well. Similar to the electron cyclotron resonance in metals, observation of the SCR would allow experimentalists to unambiguously measure the skyrmion mass which has been subject of significant controversy.

Our analytical formulas equally apply to synthetic ferrimagnets in which antiferromagnetically coupled sublattices belong to different adjacent ferromagnetic layers [52–54]. The numerical results are expected to be similar for such systems. They provide another possibility for experimental detection of the skyrmion cyclotron resonance and for obtaining the skyrmion mass.

Finally, we would like to notice the possibility of quantization of the skyrmion cyclotron motion, similar to Landau quantization of an electron's orbital motion in a magnetic field.

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